- Start with the known root predecessor.
- Check that entry.previous_hash equals the expected previous hash.
- Rebuild the exact hash payload.
- Recalculate SHA-256.
- Compare recalculated hash with entry.hash.
- Stop immediately if a mismatch occurs; otherwise continue to the end.

The guide describes this same sequential verification procedure and the tamper index reporting. ■filecite■turn1file1■L89-L107■

Simple analogy: every page of a notebook contains the fingerprint of the previous page. Change an old page and the chain no longer agrees.

Security caveat: the current code calls itself HMAC-SHA256, but it actually calls hashlib.sha256. The secret_key is stored but never used. It is therefore a plain hash chain, not an HMAC implementation.

Also, a hash chain is tamper-evident only when the trusted root/checkpoint and storage are protected. Production security needs protected keys, access control, append-only or secure storage, and preferably external anchoring or signed checkpoints.

4. MVEMPhysicsEngine — Physics Baseline

MVEM means **Mean Value Engine Model**. Instead of simulating every combustion event at crank-angle resolution, it works with averaged engine quantities. This makes it computationally attractive for real-time monitoring.

The class contains atmospheric constants, an ISA density calculator and a function that predicts healthy expected sensor values.

4.1 ISA air density

$$\rho = P / (R \cdot T)$$

The code converts altitude from feet to metres and ambient temperature from °C to kelvin. It estimates pressure using a standard-atmosphere relationship and then applies the ideal-gas relation. Density matters because engine breathing and therefore expected engine behavior change with altitude and temperature.

4.2 Density ratio

$$\text{density_ratio} = \rho / 1.225$$

1.225 kg/m³ is used as the sea-level reference density. A lower ratio represents thinner air.

4.3 Expected MAP

$$\text{exp_map} = 30 + (\text{throttle}/100) \times 70 \times \text{density_ratio}$$

MAP means Manifold Absolute Pressure. More throttle increases expected MAP; reduced air density reduces it.

4.4 Volumetric efficiency

$$\eta\blacksquare = 0.82 + 0.12(\text{RPM}/5800) - 0.05(\text{RPM}/5800)^2$$

Volumetric efficiency approximates how effectively the cylinders fill. The quadratic term introduces diminishing benefit at higher RPM. The supplied guide explicitly identifies this as the breathing correction. ■filecite■turn2file0■L337-L339■

4.5 Expected engine quantities

Quantity	What the formula captures
EGT	Ambient temperature + RPM effect + throttle effect
CHT	Ambient temperature + RPM effect + manifold-pressure effect
Oil pressure	RPM and ambient-temperature effects
Oil temperature	Ambient temperature + RPM effect
Fuel flow	Throttle + density + volumetric efficiency
Vibration	RPM-dependent quadratic increase

Key interpretation: these equations are best presented as a fast, calibrated healthy-reference model, not as a full high-fidelity thermodynamic simulation.

5. EKFStateEstimator — State Smoothing and Residuals

This class combines the MVEM expected values with actual sensor measurements and produces two useful outputs: a smoothed state estimate and physics residuals.

Important: despite the class name, the current implementation is **not a full Extended Kalman Filter**. It uses fixed-factor exponential smoothing. The project guide describes this stage as EKF-style smoothing/residual extraction. ■filecite■turn2file0■L341-L344■

5.1 What a real EKF normally has

- State vector x representing hidden engine variables.
- Nonlinear transition model f(x,u).
- Measurement model h(x).
- Covariance matrix P representing uncertainty.
- Process noise Q and measurement noise R.
- Prediction and measurement-update equations.
- Jacobian matrices for linearizing nonlinear functions.

5.2 What this code actually does

```
if self.x_est is None:
    self.x_est = obs
else:
    self.x_est = (1 - self.alpha) * self.x_est + self.alpha * obs
```

With $\alpha = 0.35$, the new observation contributes 35% and the previous estimate contributes 65%. Larger α reacts faster; smaller α produces stronger smoothing.

Example: previous EGT estimate = 700°C, new observation = 760°C → new estimate = 0.65×700 + 0.35×760 = **721°C**.

5.3 Physics residuals

ΔS = Actual − Expected

For example, ΔEGT = actual EGT − expected EGT. A positive value means the engine is hotter than the healthy reference. The code creates six residuals: EGT, CHT, oil pressure, oil temperature, fuel flow and vibration.

Residuals are the bridge between physics and AI: the AI receives both raw operating measurements and deviations from expected behavior.

6. EngineHealthEngine — Sensor Health and EHI

This module answers two separate questions: **Can I trust the sensors?** and **How healthy is the engine?**

6.1 Sensor-health logic

Sensor	Main check
EGT	Residual magnitude + high-temperature plausibility
CHT	Residual magnitude + high-temperature plausibility
Oil pressure	Residual magnitude + 0.1–10 bar plausibility
Oil temperature	Simple 20–160°C range check
Fuel flow	Residual magnitude
Vibration	Simple <15 mm/s validity check

Cross-sensor logic is useful. If EGT looks highly suspicious while CHT remains healthy, the system can suspect an EGT sensor problem rather than immediately declaring an engine fault. The guide explicitly describes this type of inconsistency test. ■filecite■turn2file0■L13-L20■

6.2 Five-pillar Engine Health Index

Subsystem	Formula idea	Meaning
Combustion	100 – 0.4|ΔEGT| – 4|ΔFuel|	Mixture, spark, burn quality
Thermal	100 – 0.7|ΔCHT| – 0.3|ΔEGT|	Cooling and thermal stress
Lubrication	100 – 22|ΔOilP| – 0.8|ΔOilT|	Oil film / bearing condition
Mechanical	100 – 14|ΔVib| – 15 if |ΔCHT|>30	Mechanical integrity
Electrical	100 – 18×max(0,27.5–Vbattery)	Power-system condition

Overall EHI = 0.25 Combustion + 0.25 Thermal + 0.20 Lubrication + 0.20 Mechanical + 0.10 Electrical.

The supplied guide uses Healthy ≥85, Normal/Slight Degradation 70–85, Degraded/Warning 50–70 and Critical <50. ■filecite■turn2file0■L39-L45■

EHI vs Twin Confidence: EHI is about the engine's condition. Twin Confidence is about how much the system should trust the current measurements/diagnosis.

7. AIDiagnosticsSuite — AI Intelligence Layer

The AI receives a 14-dimensional feature vector: **8 raw values + 6 physics residuals**. The guide documents this exact feature design. ■filecite■turn2file0■L61-L70■

7.1 Isolation Forest — “Is this unusual?”

- Unsupervised anomaly detector; training does not require every fault to be labeled.
- Unusual observations are easier to isolate using random partitions.
- The code trains it on normal telemetry.
- The raw decision_function output is transformed into a custom 0–1 anomaly score.
- anomaly_score > 0.45 sets is_anomaly = True.

The 0–1 value is a project-specific normalized anomaly score, not a universal probability that the engine will fail.

7.2 XGBoost classifier — “Which fault is it?”

ID	Fault	Typical interpretation
0	Normal Operation	No modeled fault
1	Cylinder Misfire	Incomplete combustion / ignition issue
2	Injector Coking / Lean Fuel	Restricted injector / lean mixture
3	Lubrication System Loss	Oil-pressure / lubrication degradation
4	Sensor Drift / Bias	Measurement-system fault
5	Combustion Instability	Unstable combustion
6	Thermal Overheating	Excess thermal condition
7	Abnormal Mechanical Vibration	Mechanical imbalance / vibration

XGBoost is a gradient-boosted decision-tree ensemble. Trees are built sequentially so later trees focus on errors left by earlier trees. It is well suited to structured/tabular telemetry features.

7.3 SHAP — “Why did the model say that?”

SHAP (SHapley Additive exPlanations) assigns contribution values to features. A large absolute SHAP value means that feature strongly influenced the selected model output. The code ranks the contributions and returns the top three features plus their actual values.

Example interpretation: if ΔEGT and ΔFuel have large contributions, the operator can see that the diagnosis was not produced from a mysterious black box; the model is responding strongly to measurable evidence.

7.4 XGBoost RUL regressor — “How much useful life remains?”

RUL means **Remaining Useful Life**. The regressor maps the feature vector to estimated remaining operating hours. The code then creates an uncertainty band using max(12 hours, 18% of predicted RUL).

Example: RUL = 350 h → margin = 63 h → interval = [287, 413] h.

The project guide calls this an 80% confidence interval, but the current implementation does not statistically calibrate an 80% interval; it applies a fixed heuristic margin. A production claim should be validated/calibrated on independent run-to-failure data. ■filecite■turn2file0■L97-L107■

8. TwinVisualizationAdapter — Unified Digital-Twin State

The adapter is a presentation/data-contract layer. It does not perform the core diagnosis. It collects telemetry, expected values, residuals, health scores, AI results, RUL and mission-reliability information into one dictionary that the UI can consume.

The guide describes this unified state as containing telemetry, attitude, GPS, residuals, sensor details, subsystem health, AI diagnostics, RUL and mission reliability. ■filecite■turn1file2■L164-L196■

Why this architecture is useful

- The dashboard gets one consistent data structure instead of knowing the internals of every backend module.
- Actual, expected, residual and health can be displayed together.
- The same state can drive KPI cards, charts, maps and 3D visualization.
- Backend algorithms can change without rewriting the whole front end.

The code synthesizes pitch, roll, yaw and latitude/longitude using mathematical functions of timestamp/throttle. These values are suitable for an MVP visualization, but they are **not real navigation or flight-control estimates**.

Sensor markers are intended to be spatially meaningful: a maintainer can select a sensor and see actual value, expected value, residual, unit and health.

9. End-to-End Worked Example

Assume a telemetry frame arrives with:

```
RPM = 4800
Throttle = 65%
Altitude = 10,000 ft
Ambient temperature = 10°C
Actual EGT = 760°C
Actual CHT = 180°C
Actual oil pressure = 3.7 bar
Actual fuel flow = 14.0 L/h
Actual vibration = 2.8 mm/s
```

Step 1 — Physics

MVEM uses the operating condition to calculate an expected healthy state. Suppose expected EGT is 680°C.

ΔEGT = 760 − 680 = +80°C

Step 2 — State estimate

The smoothing stage blends the current observation with the previous estimate, reducing the effect of random sensor noise.

Step 3 — Health

Residuals reduce the relevant sensor and subsystem health scores. Large EGT deviation affects combustion and thermal health.

Step 4 — AI

The 14 features are passed to Isolation Forest, XGBoost classifier, SHAP and RUL regressor. A hypothetical result could be anomaly = 0.72, fault = Injector Coking, high classifier confidence, top SHAP contributors = ΔEGT/ΔFuel/CHT, and reduced RUL.

Step 5 — Audit

The event is serialized, combined with the previous hash and hashed. The new digest becomes the predecessor for the next event.

Step 6 — Visualization

The adapter packages the complete state. The operator can see EHI, Twin Confidence, anomaly score, diagnosis, SHAP evidence, RUL and sensor-level evidence in one place.

10. Mathematical Concepts You Should Know

Concept	Core equation / idea	Purpose
Ideal gas law	ρ = P/(RT)	Air-density estimation
Density ratio	ρ/ρ■	Altitude/weather correction
Volumetric efficiency	η■(RPM)	Approximate engine breathing
Residual	Δ = actual − expected	Deviation from healthy physics
Smoothing	x■=(1−α)x■■■+αy■	Noise reduction
EHI	Σw■H■	Overall engine health
Isolation Forest	Isolation depth → unusualness	Anomaly detection
XGBoost	Boosted decision trees	Fault classification / RUL regression
SHAP	Feature contribution values	Explain AI output
Hash chain	H■=SHA256(H■■■|data)	Tamper detection

Mental model: Physics gives the baseline → smoothing reduces noise → residuals expose deviations → health scoring summarizes condition → ML recognizes complex patterns → SHAP provides evidence → RUL forecasts remaining margin → cryptography protects the recorded evidence.

11. Important Implementation Corrections and Caveats

Audit says HMAC-SHA256, but code uses plain SHA-256.

hashlib.sha256(...) is used. The secret_key is never used. A true HMAC implementation would use the hmac module and a secret key.

EKFStateEstimator is not a full EKF.

It uses exponential smoothing. There is no covariance P, Q/R matrices, Jacobian or prediction/update cycle.

RUL uncertainty is heuristic.

±18% with a minimum of 12 hours is not automatically an 80% statistically calibrated interval.

Hash chain is tamper-evident, not automatically immutable.

The trusted root/checkpoint and storage must be protected from an attacker who can rewrite the entire dataset.

Visualization sensor-key mismatch.

The health engine returns keys such as egt_sensor/cht_sensor, while the adapter looks for EGT/CHT/OilP/Fuel/Vib. This can cause fallback health values instead of the calculated scores.

exp_rpm is not returned by MVEM.

The adapter requests mvem_exp['exp_rpm'], but the physics engine does not create that field, so the fallback actual RPM is used.

Attitude and GPS are synthesized.

They are demo trajectories, not real IMU/GNSS/navigation fusion.

Physics formulas are MVP/calibrated relationships.

They should not be described as a validated high-fidelity thermodynamic model without engine-specific calibration and validation.

Training data are synthetic.

The AI is initially trained from generated telemetry and run-to-failure trajectories. Real flight/lab data are needed to establish deployment performance.

These are engineering boundaries, not reasons to discard the design. They show what must be strengthened when moving from a hackathon MVP toward a validated production/flight system.

12. What Each Output Means to an Operator

Output	Question answered	Meaning
MVEM expected	What should a healthy engine look like now?	Physics baseline
Residual ΔS	How far is reality from that baseline?	Deviation/evidence
Sensor score	Can I trust this sensor?	Measurement quality
Twin Confidence	How much should I trust the current twin diagnosis?	Diagnostic trust
Subsystem health	Which area is deteriorating?	Localized condition
EHI	How healthy is the engine overall?	0–100 summary
Anomaly score	Is the current pattern unusual?	Novelty score
Fault class	What fault is most likely?	Diagnosis
Fault confidence	How strongly does the classifier support it?	Model confidence
SHAP contributors	Why did the model choose it?	Evidence
RUL	How much useful life is estimated to remain?	Forecast
RUL interval	How uncertain is that forecast?	Prediction band
Audit result	Has the recorded chain stayed consistent?	Integrity evidence

The strongest presentation story is not “AI says there is a fault.” It is: **physics establishes what should happen; residuals show what is abnormal; health scoring localizes the degradation; AI identifies the likely fault; SHAP shows why; RUL estimates the remaining margin; and the audit chain preserves the evidence trail.**

13. Viva / Presentation Questions

Q: Why combine physics and AI?

A: Physics provides an interpretable healthy baseline and reduces dependence on purely data-driven pattern matching.

Q: Why are residuals important?

A: They express deviation from expected behavior after accounting for operating conditions.

Q: Difference between EHI and Twin Confidence?

A: EHI measures engine condition; confidence measures trust in the sensing/diagnostic result.

Q: Why Isolation Forest?

A: It detects unusual patterns without requiring labels for every possible fault.

Q: Why XGBoost?

A: It is strong for nonlinear structured/tabular data and supports both classification and regression.

Q: Why SHAP?

A: It makes the diagnosis interpretable by showing which features contributed most.

Q: What is RUL?

A: Remaining Useful Life: estimated useful operating time before a defined degradation/failure endpoint.

Q: Why a cryptographic chain?

A: It makes modification of historical records detectable through broken hash continuity.

Q: Is the current EKF a true EKF?

A: No. It is fixed-factor exponential smoothing; a true EKF would maintain uncertainty/covariance and perform prediction/update calculations.

Q: Is the current RUL band a validated 80% interval?

A: No. It is a heuristic ±18% band with a minimum width and requires calibration/validation before a formal confidence claim.

Final Takeaway

AeroTwin-PX is a **hybrid engine condition-monitoring pipeline**. The physics layer establishes expectations; the residual layer exposes deviations; the health engine turns deviations into interpretable health; the AI layer detects, diagnoses and forecasts; SHAP explains the diagnosis; the audit layer protects the evidence trail; and the visualization adapter turns everything into an operator-facing digital-twin state.

The supporting project guide documents the same seven-module architecture, health model, AI stack, audit chain and visualization contract. ■filecite■turn1file7■L537-L565■